The documentation and process conversion measures necessary to comply with this revision shall be completed by 3 May 1994

INCH-POUND

MIL-S-19500/368C
3 January 1994
SUPERSEDING
MIL-S-19500/368B
4 April 1990

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPES 2N3439, 2N3439L, 2N3440, AND 2N3440L
JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, low-power, high voltage transistors. Three levels of product assurance are provided for each encapsulated device type as specified in MIL-S-19500, and two levels of product assurance for each unencapsulated device type die.

1.2 Physical dimensions. See figures 1 (similar to TO-39 and TO-5) and 2 (JANHC and JANKC).

1.3 Maximum ratings.

Type	P_T 1/ T_A = +25°C 2/	P_T 3/ T_C = +25°C	V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_J and T_{STG}
	W	W	V dc	V dc	V dc	A dc	°C
2N3439	0.8	5	450	7	350	1	-55 to +200
2N3440	0.8	5	300	7	250	1	-55 to +200

1/ Derate linearly, 4.57 mW/°C for T_A > +25°C.
2/ T_A = room ambient as defined in the general requirements of MIL-STD-750.
3/ Derate linearly, 28.5 mW/°C for T_C > +25°C.

1.4 Primary electrical characteristics.

	h_{FE2} 1/	h_{FE1} 1/	$\|h_{fe}\|$	C_{obo}	$V_{BE(sat)}$ 1/	$V_{CE(sat)}$
	V_{CE} = 10 V dc I_C = 2 mA dc	V_{CE} = 10 V dc I_C = 20 mA dc	V_{CE} = 10 V dc I_C = 10 mA dc f = 5 MHz	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	I_C = 50 mA dc I_B = 4 mA dc	I_C = 50 mA dc I_B = 4 mA dc
				pF	V dc	V dc
Min	30	40	3			
Max		160	15	10	1.3	0.5

1/ Pulsed, (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronic Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4, Section D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

JANHC . . . . . High reliability product assurance level for unencapsulated devices.
JANKC . . . . . Space reliability product assurance level for unencapsulated devices.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core is permitted. Lead finish shall be gold plated or solder dipped. Where a choice of lead material or finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.4.1 Marking of JANHC and JANKC die. For JANHC and JANKC die, the following marking shall be used (example):

JANHCAM2N3439
↑↑↑↑
|||└ RHA level (see MIL-S-19500).
||└── Source of manufacturer (see figure 3).
|└─── Unencapsulated.
└──── Product assurance level 1/.

1/ Two levels of product assurance levels are provided for unencapsulated devices, H and K (see MIL-S-19500.

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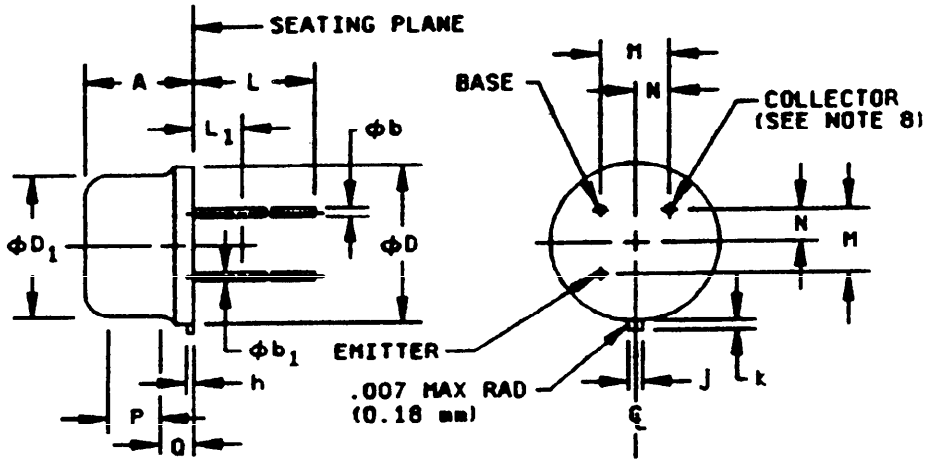


FIGURE 1. Physical dimensions.

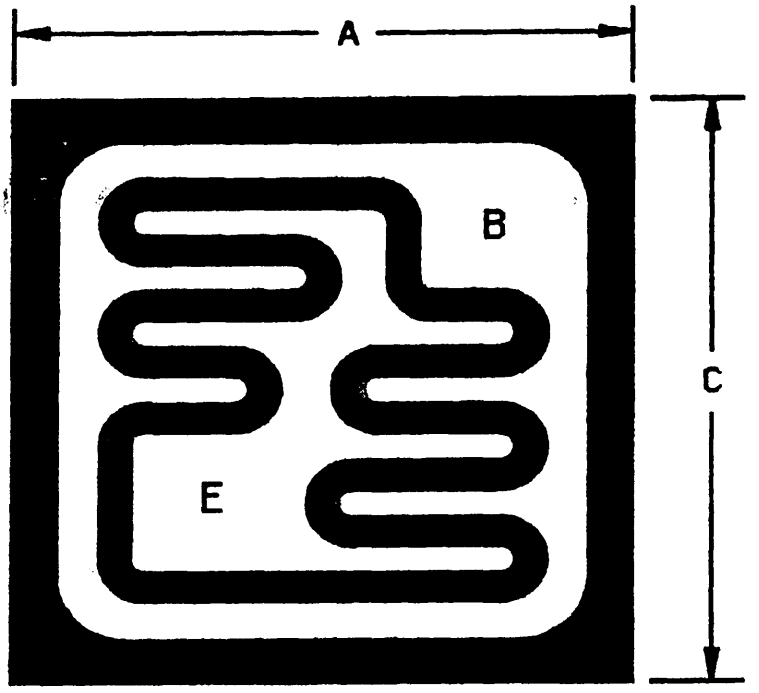
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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.240	.260	6.10	6.60	
ϕb	.016	.021	0.41	0.53	3, 10
ϕb_1	.016	.019	0.41	0.48	4, 10
ϕD	.335	.370	8.51	9.40	
ϕD_1	.305	.335	7.75	8.51	
K	.029	.045	0.74	1.14	9
h	.009	.125	0.23	3.18	
j	.028	.034	0.71	0.86	
L	See notes 10, 12, and 13				
L_1	---	.050	---	1.27	11
M	.1414 Nom		3.591 Nom		7
N	.0707 Nom		1.796 Nom		7
P	.100	---	2.54	---	5
Q	---	---	---	---	6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond 0.250 inch (6.35 mm) from the seating plane.
4. Measured in the zone 0.50 inch (1.27 mm) from the seating plane.
5. Variations on dimension ϕD_1 in this zone shall not exceed 0.010 inch (0.25 mm).
6. Outline in this zone is not controlled.
7. When measured in a gauging plane 0.054 +0.001, -0.000 inch (1.37 +0.03, - 0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within 0.007 inch (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 3 shows the preferred measured method.
8. The collector shall be electrically connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All three leads (see 3.3.1).
11. Diameter of leads in this zone is not controlled.
12. For transistor types 2N3439 and 2N3440, L = 0.500 inch (12.70 mm) minimum, and 0.750 inch (19.05 mm) maximum.
13. For transistor types 2N3439L and 2N3440L, L = 1.500 inches (38.10 mm) minimum, and 1.750 inches (44.45 mm) maximum.

FIGURE 1. Physical dimensions - Continued.

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A version

Ltr	Inches		Millimeters	
	Min	Max	Min	Max
A	.038	.044	0.97	1.12
C	.038	.044	0.97	1.12

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die thickness are, 0.006 inch (0.15 mm) to 0.012 inch (0.30 mm).
   Metallization, top: Al = 17,500 Å minimum, 20,000 Å nominal.
   Back: Au = 2,500 Å minimum, 3,000 Å nominal.
   Bonding pad: B = 0.004 inch (0.10 mm) by 0.005 inch (0.13 mm).
   E = 0.004 inch (0.10 mm) by 0.005 inch (0.13 mm).
4. Backside is collector.
5. Requirements in accordance with MIL-S-19500 (appendix H) are performed in a TO-5 package.

FIGURE 2. Physical dimensions JANHC and JANKC (die).

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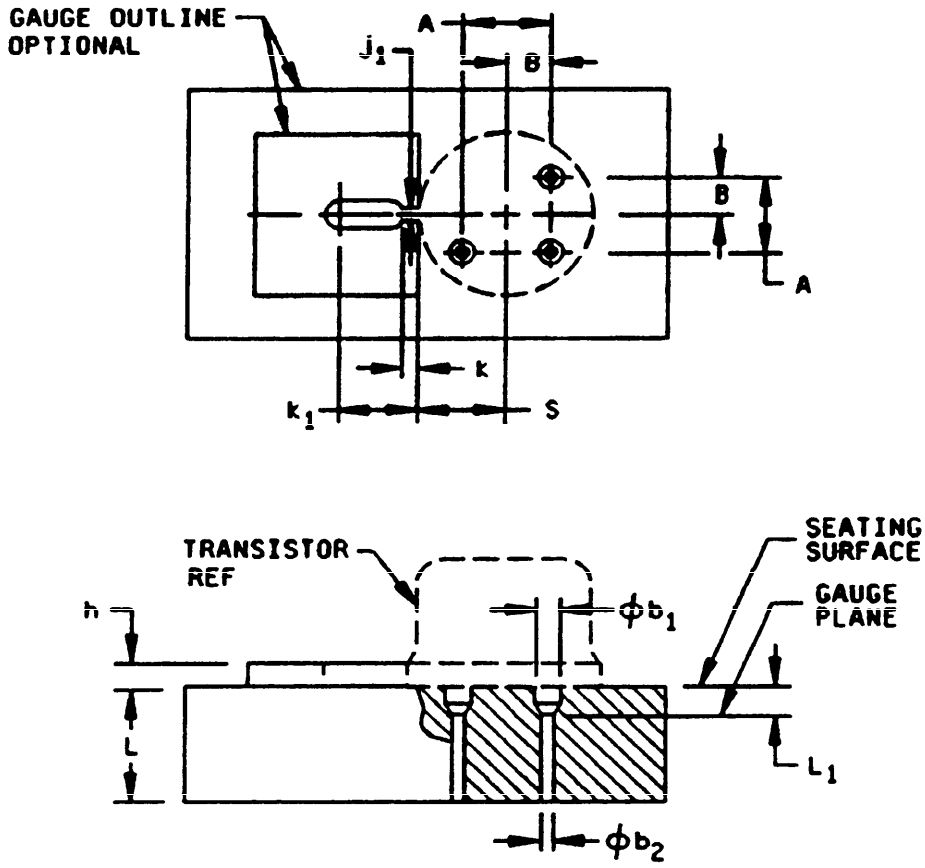


FIGURE 3. Gauge for lead and tab location.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	0.1409	0.1419	3.579	3.604	
B	0.0702	0.0712	1.783	1.808	
ϕb1	0.0595	0.0605	1.511	1.537	
ϕb2	0.0325	0.0335	0.8551	0.851	
h	0.150 Nominal		3.81 Nominal		
j1	0.0350	0.0355	0.889	0.902	
K	0.009	0.011	0.23	0.28	
K1	0.125 Nominal		3.18 Nominal		
L	0.372	0.378	9.45	9.60	
L1	0.054	0.055	1.37	1.40	
s	0.182	0.199	4.62	5.05	3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The location of the tab locator within the limits of dimension S will be determined by the tab and flange dimensions of the device being checked.
4. The following gauging procedure shall be used:
   The use of a pin straightener prior to insertion in the gauge is permitted. The device being measured shall be inserted until it's seating plane is 0.125 inch (3.18 mm) ±0.010 inch (0.2 mm) from the seating surface of the gauge. A spacer may be used to obtain the 0.125 inch (3.18 mm) distance from the gauge seat prior to force application. A force of 8 ±0.5 ounces shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge.

FIGURE 3. Gauge for lead and tab location - Continued.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTXV and JANTX levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CEO1} and h_{FE1}	Not applicable
11	I_{CEO1}; h_{FE1}; ΔI_{CEO1} = 100 percent or 0.5 μA dc, whichever is greater; Δh_{FE1} = ±15 percent change of initial value.	I_{CEO1} and h_{FE1}
12	See 4.3.2	See 4.3.2
13	Subgroups 2 and 3 of table I herein; ΔI_{CEO1} = 100 percent or 0.5 μA dc, whichever is greater; Δh_{FE1} = ±20 percent change of initial value.	Subgroup 2 of table I herein; ΔI_{CEO1} = 100 percent or 0.5 μA dc whichever is greater; Δh_{FE1} = ±20 percent change of initial value.

4.3.1 Screening (JANHC and JANKC die). Screening of JANHC and JANKC die shall be in accordance with MIL-S-19500, appendix H. As a minimum, die shall be 100 percent probed. Test limits and conditions shall be chosen by the manufacturer to demonstrate compliance with electrical characteristics specified in table I herein.

4.3.2 Power burn-in. Power burn-in conditions are as follows:

P_T = 800 mW dc; V_{CB} = 200 V dc; T_A = room ambient as defined in the general requirements of MIL-STD-750.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa and IVb (JANS, JANTX and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

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4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Conditions
B4	1037	V_{CB} = 200 V dc; P_T = 800 mW at T_A = room ambient as defined in the general requirements of MIL-STD-750; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles.
B5	1027	V_{CB} = 200 V dc; 168 hours, P_T = 915 mW at T_A = 25°C or adjusted as required according to the chosen T_A to give and average T_J = 225°C.
B6	3131	See 4.5.2.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Conditions
B3	1027	V_{CB} = 200 V dc; P_T = 800 mW at T_A = room ambient as defined in the general requirements of MIL-STD-750. No heat sink or forced-air cooling on the devices shall be permitted.
B5	3131	See 4.5.2

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Conditions
C2	2036	Test condition E.
C6	1026	V_{CB} = 200 V dc; P_T = 800 mW at T_A = room ambient as defined in the general requirements of MIL-STD-750. No heat sink or forced-air cooling on device shall be permitted.

4.5 Methods of inspection. Methods of inspection and test shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131, MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 150 mA dc for $R_{\theta JC}$.

b. Collector to emitter voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be +25°C ≤ T_R ≤ +75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to cas for $R_{\theta JC}$.

f. Maximum limit of $R_{\theta JC}$ shall be 35°C/W.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector to emitter cutoff current	3041	Bias condition D	I_{CEO}		2	µA dc
2N3439		V_{CE} = 300 V dc				
2N3440		V_{CE} = 200 V dc				
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 7 V dc	I_{EBO}		10	µA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX}		5	µA dc
2N3439		V_{CE} = 450 V dc				
2N3440		V_{CE} = 300 V dc				
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}		2	µA dc
2N3439		V_{CB} = 360 V dc				
2N3440		V_{CB} = 250 V dc				
Collector to base cutoff current	3036	Bias condition D	I_{CBO2}		5	µA dc
2N3439		V_{CB} = 450 V dc				
2N3440		V_{CB} = 300 V dc				
Base emitter voltage (nonsaturated)	3066	Test condition A, pulsed (see 4.5.1), I_C = 50 mA dc, I_B = 4 mA dc	$V_{BE(sat)}$		1.3	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1), I_C = 50 mA dc, I_B = 4 mA dc	$V_{CE(sat)}$		0.5	V dc
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 10 V dc, I_C = 20 mA dc	h_{FE1}	40	160	
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 10 V dc, I_C = 2 mA dc	h_{FE2}	30		
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 10 V dc, I_C = 0.2 mA dc	h_{FE3}	10		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D	I_{CBO3}		100	µA dc
2N3439		V_{CB} = 360 V dc				
2N3440		V_{CB} = 250 V dc				
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 10 V dc, I_C = 20 mA dc	h_{FE4}	15		
Subgroup 4						
Pulse response:	3251	Test Condition A				
Turn-on time		V_{CC} = 200 V dc, I_C = 20 mA dc, I_{B1} = 2 mA dc, see figure 5	t_{on}		1	µs
Turn-off time		V_{CC} = 200 V dc, I_C = 20 mA dc, I_{B1} = $-I_{B2}$ = 2 mA dc, see figure 5	t_{off}		10	µs
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 10 mA dc, f = 5 MHz	$\|h_{fe}\|$	3	15	
Open capacitance input open circuited	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		10	pF
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc, I_C = 5 mA dc, f = 1 kHz	h_{fe}	25		
Input capacitance (output open circuited)	3240	V_{EB} = 5 V dc, I_C = 0, 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		75	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (continuous dc)	3051	See figure 6, T_C = 25°C, 1 cycle, t = 1.0 s				
Test 1		V_{CE} = 5 V dc, I_C = 1 A dc				
Test 2 Only 2N3439		V_{CE} = 350 V dc I_C = 14 mA dc				
Test 3 Only 2N3440		V_{CE} = 250 V dc I_C = 20 mA dc				
Electrical measurements		See table II, steps 1 and 3				
Breakdown voltage, collector to emitter	3011	I_C = 50 mA dc, R_{BB1} = 100 ohms V_{BB1} = 6 V, L = 25 mH minimum f = 30 to 60 Hz, see figure 4	$V_{BR(CEO)}$			
2N3439				350		V dc
2N3440				250		V dc

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Group A, B, and C electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limit Min	Limit Max	Unit
1.	Collector to emitter cutoff current	3041	Bias condition D	I_{CEO1}			
	2N3439		V_{CE} = 300 V dc			2	µA dc
	2N3440		V_{CE} = 200 V dc			2	µA dc
2.	Collector to emitter cutoff current	3041	Bias condition D	I_{CEO2}			
	2N3439		V_{CE} = 300 V dc			4	µA dc
	2N3440		V_{CE} = 200 V dc			4	µA dc
3.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 20 mA dc, pulsed (see 4.5.1)	h_{FE1}	40	160	
4.	Collector to emitter cutoff current	3041	Bias condition D, V_{BE} = 1.5 V dc	ΔI_{CEO}		100% of initial value or 2 µA dc, whichever is greater	
	2N3439		V_{CE} = 300 V dc				
	2N3440		V_{CE} = 200 V dc				
5.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 20 mA dc, pulsed (see 4.5.1)	Δh_{FE1}		±25 percent change in initial recorded value.	
6.	Base emitter voltage (saturated)	3066	Test condition A, pulsed (see 4.5.1), I_C = 50 mA dc, I_B = 4 mA dc	$V_{BE(sat)2}$		1.4	V dc
7.	Collector to emitter voltage (saturated)	3071	I_C = 50 mA dc, I_B = 4 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.6	V dc
8.	Collector to emitter voltage (saturated)	3071	I_C = 50 mA dc, I_B = 4 mA dc, pulsed (see 4.5.1)	$\Delta V_{CE(sat)2}$		50 mV dc from initial value.	

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 3, 6, and 7.
b. Subgroup 4, see table II herein, steps 1, 3, 6, 7, and 8.
c. Subgroup 5, see table II herein, steps 1, 3, 4, 5, 6, and 7.

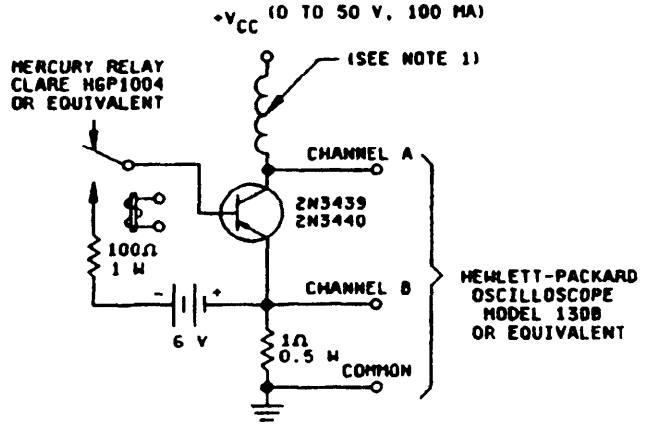
2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

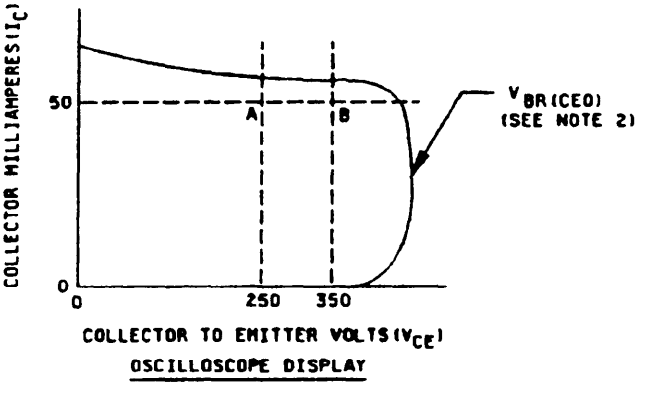
a. Subgroup 2, see table II herein, steps 1 and 3.
b. Subgroups 3 and 6, see table II herein, steps 2, 3,and 5.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table II herein, steps 1 and 3.
b. Subgroup 6, see table II herein, steps 2, 3, and 5.

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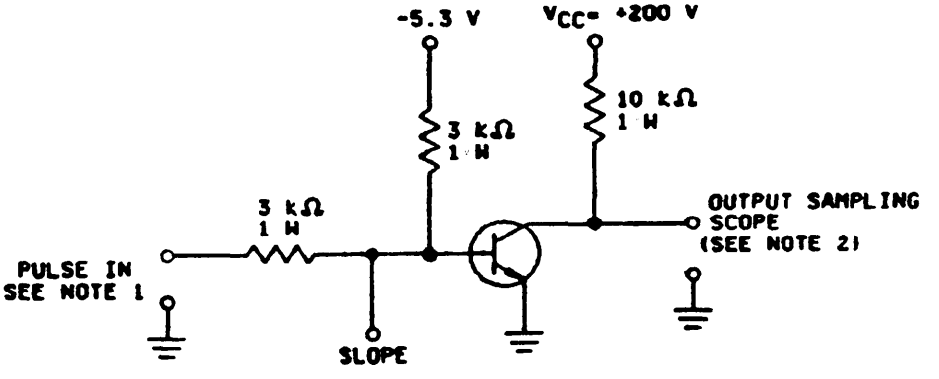


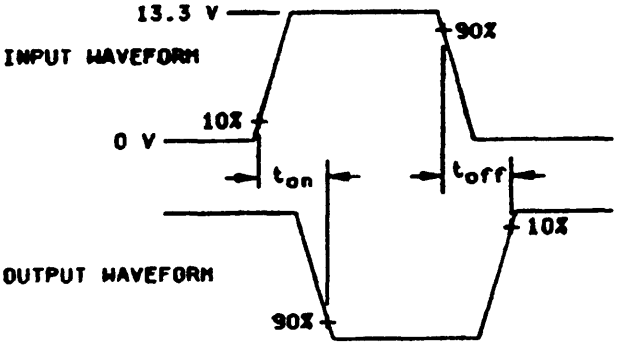
OSCILLOSCOPE DISPLAY

NOTES:
1. L ≥ 25 mH, Q = 76, R = 83.4 ohms, I = 75 mA (J. W. Miller number 6308 or equivalent).
2. $V_{BR(CEO)}$ is acceptable when the trace falls to the right and above point "A" for type 2N3440. The trace shall fall to the right and above point "B" for type 2N3439.

FIGURE 4. $V_{BR(CEO)}$ measurement test circuit.

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NOTES:
1. The rise time (t_r) and fall time (t_f) of applied pulse shall be ≤ 20 ns, duty cycle ≤ 2 percent, generator source impedance shall be 50 ohms, pulse width = 20 µs.
2. Output sampling oscilloscope: Z_{IN} ≥ 100 kohms, C_{IN} ≤ 50 pF, and rise time ≤ 1.0 µs.

FIGURE 5. Pulse response test circuit.

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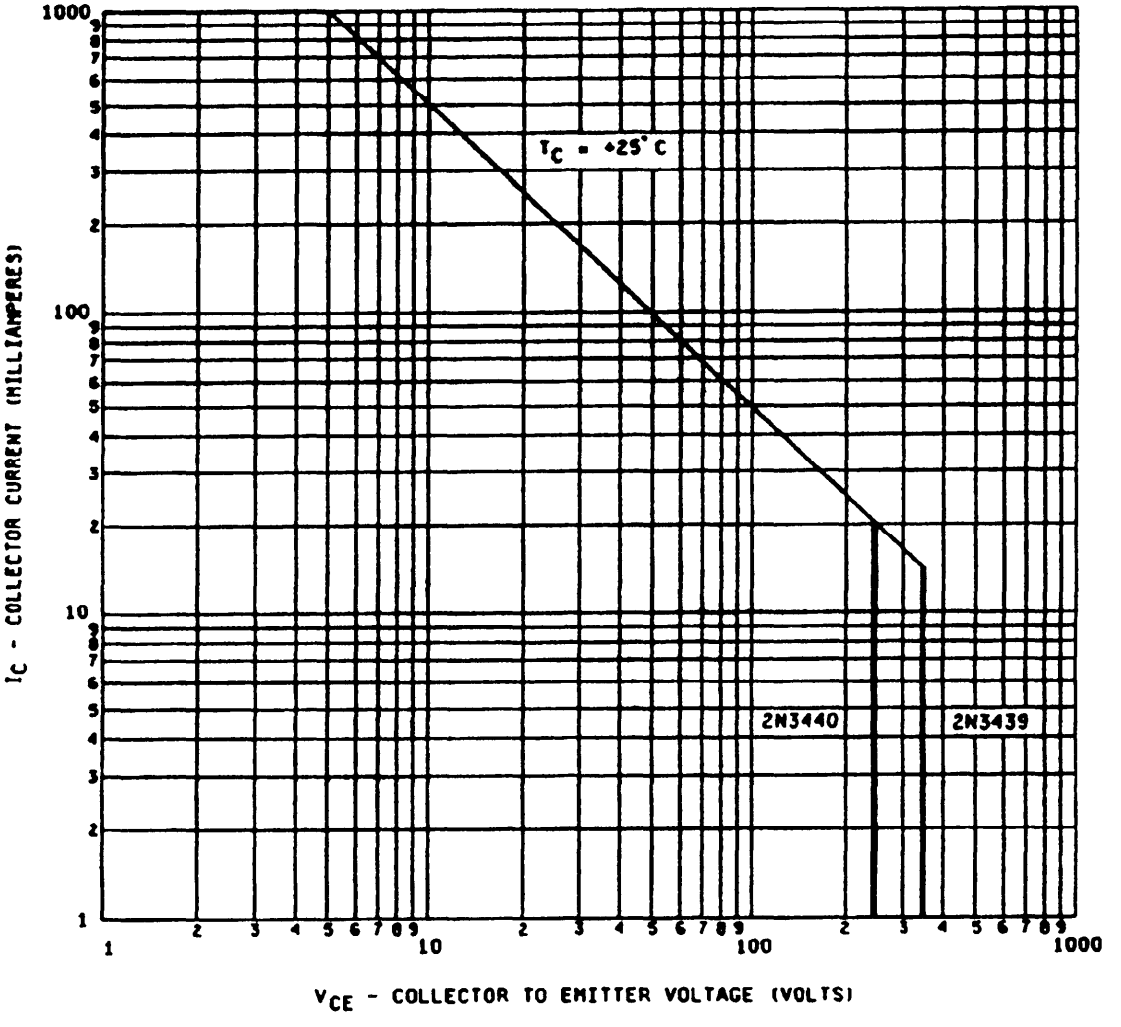


FIGURE 6. Maximum safe operating graph (continuous dc).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead material and finish as specified (see 3.3.1).

c. Type designation, product assurance level and for die acquisition, the JANHC or JANKC letter version should be specified (see figure 2).

6.3 Supersession information. Devices covered by this specification are a direct substitute for the manufacturer's and user's PIN. This information in no way implies that manufacturer's PIN's are suitable as a substitute for the military Part or Identifying Number (PIN). The term PIN is equivalent to the term (part number, identification number, and type designator) which was previously used in this specification.

Quality level	Military PIN	Manufacturer's CAGE code	Manufacturer's and user's part number
JAN	2N3439/40	18722 04713	2N2329/40 ST1577H
JTX	2N3439/40	18722 04713	2N229/40 ST1114H ST1366H ST1511H ST1721H ST830H ST982H11 ST1542H1 ST1542H2
JTXV	2N3439/40	18722 04713	2N3439/40 ST7046H

6.4 Suppliers of JANC die. The qualified JANC suppliers with the applicable letter version (example JANHCA2N3439) will be identified on the QPL.

JANC ordering information	
	Manufacturer
PIN	33178
2N3439	JANHCA2N3439
2N3440	JANHCA2N3440
2N3439	JANKCA2N3439
2N3440	JANKCA2N3440

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - MI, SM
Navy - AS, CG, MC
Air Force - 15, 19, 85, 99
DLA - ES

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5961-1541)

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